

Problem

Design a problem to help other students better understand source-free RLC circuits.

Step-by-step solution

Step 1 of 4

Consider the source-free series RLC circuit.

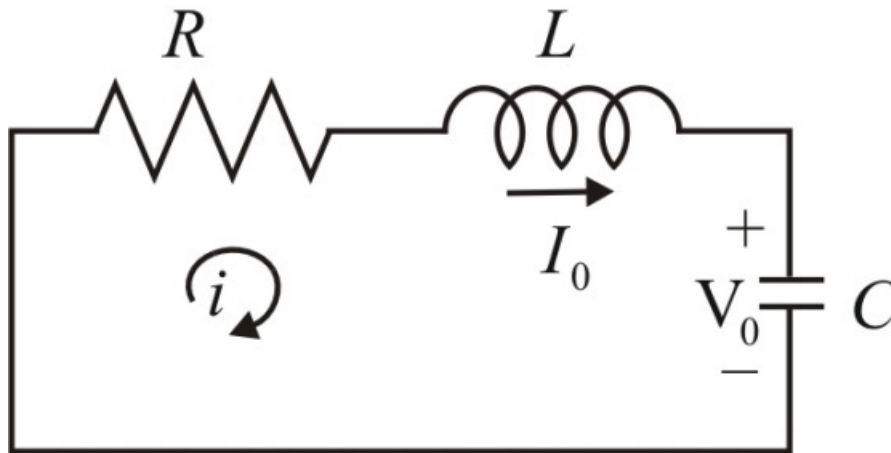


Figure 1

Step 2 of 4

Let the initial capacitor voltage is, V_0 and the initial inductor current is I_0 .

That is,

$$v_C(0) = V_0$$

$$i_L(0) = I_0$$

Write the Kirchhoff's voltage law for the loop in Figure 1.

$$Ri + L \frac{di}{dt} + \frac{1}{C} \int_{-\infty}^t i dt = 0$$

Differentiate with respect to ' t ' on both sides.

$$R \frac{di}{dt} + L \frac{d^2i}{dt^2} + \frac{i}{C} = 0$$

$$\frac{d^2i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = 0$$

Step 3 of 4

To solve such a second order differential equation requires initial conditions.

At $t = 0$,

$$Ri_c(0) + L \frac{di(0)}{dt} + v_c(0) = 0$$

$$\frac{di(0)}{dt} = \frac{1}{L} [-Ri_c(0) - v_c(0)]$$

$$\frac{di(0)}{dt} = -\frac{1}{L} (RI_0 + V_0)$$

The characteristic equation from differential equation is,

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

Calculate the roots of the equation.

$$s_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

(or)

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Where,

$$\alpha = \frac{R}{2L}$$

And,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Step 4 of 4

There are three types of solutions,

(1)

If $\alpha > \omega_0$, the response is over damped.

The solution is,

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

(2)

If $\alpha = \omega_0$, the response is critically damped.

The solution is,

$$i(t) = (A_1 + A_2 t) e^{-\alpha t}$$

(3)

If $\alpha < \omega_0$, the response is under damped.

The solution is,

$$i(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

Where,

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$